

PARA [para_000]: Do Language Embeddings Capture...

Do Language Embeddings Capture Scales?

PARA [para_001]: Xikun Zhang^{*} Stanford Univers...

PARA [para_003]: Deepak Ramachandran^{*} Google Re...

Xikun Zhang^{*}

Deepak Ramachandran^{*}

Stanford University

Google Research

xikunz2@cs.stanford.edu

ramachandrand@google.com

PARA [para_005]: Ian Tenney Google Research

PARA [para_007]: Yanai Elazar

PARA [para_010]: Dan Roth University of Pennsyl...

Ian Tenney

Yanai Elazar

Dan Roth

Google Research

Bar Ilan University, AI2

University of Pennsylvania

iftenney@google.com

yanaiela@gmail.com

danroth@seas.upenn.edu

BEAT [para_012]: Abstract

Abstract

VISUAL: figure: The figure illustrates a machi...

COMMENT [ann_24]: Predicted dis

PARA [para_064]: 2020

Pretrained Language Models (LMs) have been shown to possess significant linguistic, common sense and factual knowledge. One form

HIGHLIGHT [ann_07]: e. One form

HIGHLIGHT [ann_08]: of knowledge that has not been studied yet

PARA [para_063]: Oct

HIGHLIGHT [ann_20]: magnitudes of objects.

PARA [para_062]: 11

in this context is information about the scalar magnitudes of objects. We show that pre-trained language models capture a significant amount of this information but are short of the capability required for general common-sense reasoning. We identify contextual information in pre-training and numeracy as two key factors affecting their performance, and show that a simple method of canonicalizing numbers can have a significant effect on the results. ¹

BEAT [para_012]: 1. Introduction

1. Introduction

PARA [para_060]: arXiv:2010.05345v1

PARA [para_015]: The success of contextualized ...

The success of contextualized pretrained Language Models like BERT (Devlin et al., 2018) and ELMo (Peters et al., 2018) on tasks like Question Answering and Natural Language Inference, has led to speculation that they are good at (Especially with Reasoning (CSR)).

On one hand, recent work has approached this question by measuring the ability of LMs to answer questions about physical common sense (Bisk et al., 2020) (“How to separate egg whites from yolks?”), temporal reasoning (Zhou et al., 2020) (“How long does a basketball game take?”), and spatial reasoning (Zhou et al., 2020) (“How to separate egg whites from yolks?”), and other hand, after realizing some high-level reasoning skills like this may be difficult to learn from a language modeling objective only, (Geva et al., 2020) injects numerical reasoning skills into LMs by additional pretraining on a

dataset. All of these skills are prerequisites for CSR.

Both authors contributed equally.

Work done during an internship at Google Research.

Code and models are available at <https://github.com/google-research-datasets/numbert.>

Aims to tackle a task: predicting scalar attribute...

Our analysis shows that contextual encoders, like BERT and ELMo, perform better than noncontextual ones, like Word2Vec, on scalar probing de-

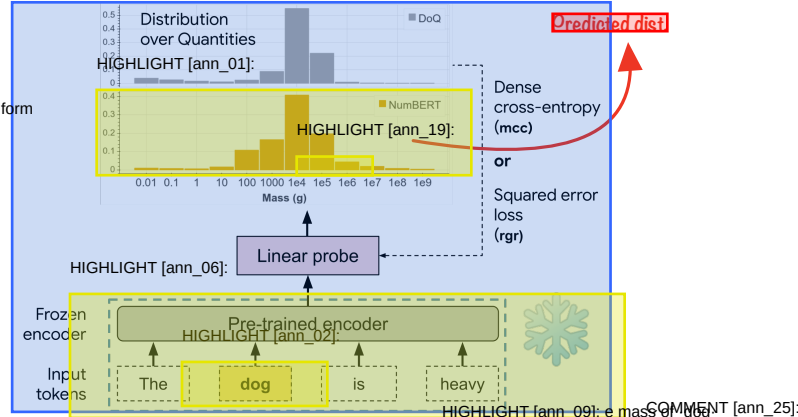


Figure 1: Scalar probing example. The mass of “dog” is a distribution (gray histogram) concentrated around 10-100kg. We train a linear model over a frozen (shown by the snowflake in the figure) encoder to predict this distribution (orange histogram) using either a dense cross-entropy or a regression loss (Section 3).

PARA [para_039]: Here, we address a simpler tas...

Here, we address a simpler task which is another pre-requisite for CSR. The prediction of scalar attributes, a task we call *Scalar Probing*. Given an object (such as a “wedding ring”) and an attribute with continuous numeric values (such as Mass or Price), can an LM’s representation of the object predict the value of that attribute? Since in general, there may not be a single correct value for such attributes due to polysemy (“crane” as a bird, versus construction equipment) or natural variation (e.g. different breeds of dogs), we interpret this as a task of predicting a distribution of possible values for this attribute, and compare it to a ground truth distribution of such values. An overview of this scalar probing is shown in Figure 1. Examples of ground-truth distributions and model predictions for different objects and attributes are shown in Figure 2.

Our analysis shows that contextual encoders, like BERT and ELMo, perform better than noncontextual ones, like Word2Vec, on scalar probing de-

Our analysis shows that contextual encoders, like BERT and ELMo, perform better than noncontextual ones, like Word2Vec, on scalar probing de-

Our analysis shows that contextual encoders, like BERT and ELMo, perform better than noncontextual ones, like Word2Vec, on scalar probing de-

Our analysis shows that contextual encoders, like BERT and ELMo, perform better than noncontextual ones, like Word2Vec, on scalar probing de-

Our analysis shows that contextual encoders, like BERT and ELMo, perform better than noncontextual ones, like Word2Vec, on scalar probing de-

Our analysis shows that contextual encoders, like BERT and ELMo, perform better than noncontextual ones, like Word2Vec, on scalar probing de-

COMMENT [ann_74]: Intuition says that anything that can perform well...

- COMMENT [ann_75]: If the contextual models like BERT trained from contextual tasks will perform well on non-contextual as well? But is that really true?
- COMMENT [ann_76]: It would be fair to say that BERT is more expressive than Word2Vec, is it due to the fact that the BERT has "context" aware training or is it rather just that BERT is more expressive than Word2Vec? Let's find out...

HIGHLIGHT [ann_41]: spite the task being non-contextual (Mikolov et al.,

PARA [para_130]: Animals and PRICE of Household...

HIGHLIGHT [ann_121]: By using scientific notation, we can represent numbers in pre-training can have

Animals and PRICE of Household products.

3 Probing Model

PARA [para_132]: We probe three different embed...

HIGHLIGHT [ann_146]: We probe three different embedding models:

Word2vec (Mikolov et al., 2013), ELMo (Peters et al., 2018) and BERT (Devlin et al., 2018) (the latter two of which are contextualized encoders). For each encoder, the input layer extracts an embedding of the object and the probing layer predicts the scalar magnitude.

Input representations For Word2vec, we follow the standard practice of averaging the embeddings of each word in the object's name. If an object name is a full phrase in the dictionary, we use its embedding instead. As BERT and ELMo are contextual text encoders operating on full sentences,

we generate artificial sentences with the following templates:

- MASS:** The X is heavy.
- PRICE:** The X is expensive.
- LENGTH:** The X is big.

and use the CLS token embedding (for BERT) or final state embedding (for ELMo) as the input representation. For LENGTH measurements in DoQ can be widths or heights as well. Variations of these templates with different adjectives and sentence structures (e.g. "The X is small." or "What is the length of X?" for LENGTH) led to very similar performance in our evaluations.

Probes We use linear probes in all cases following many previous probing work (Shi et al., 2016; Stinger et al., 2016; Pimentel et al., 2020) since we want to use a simple probe to find easily accessible information in a representation. Hewitt and Liang (2019) also demonstrate that linear probes achieve relatively high selectivity compared to non-linear ones like MLP.

We experiment with two different approaches for predicting scales:

Regression (rgr) For the point estimate, we use a standard Linear Regression model trained

We use Word2Vec embeddings of dimension size 500 trained on Wikipedia, BERT-Base (L=12, H=768, A=12, Total Parameters=110M) trained on Wikipedia+Books and ELMo-Small (LSTM Hidden Size=1024, Output Size=128, #Highway Layers=1, Total Parameters=13.6M) trained on the 1 Billion Word Benchmark, approximately 800M tokens of news crawl data from WMT 2011.

2 Problem Definition and Data

PARA [para_096]: We define the scalar probing task...

We define the scalar probing task (see Figure 1) as the problem of predicting a distribution over

of a scalar attribute of an object. We map these values into logarithmically spaced buckets, so that

our task is equivalent to predicting the distribution

of the order of magnitude of the target value. We

explore both models that predict the full distribution and models that predict a point estimate of the

value, which is essentially a distribution with all

the mass concentrating on one bucket.

Our primary resource for the scalar probing task

is Distributions over Quantities (DoQ, Elazar et al., 2019), which consists of empirical counts of scalar

attribute values associated with >350K nouns, adjectives, and verbs over 10 different attributes, collected from web data. In this work, we focus only

on nouns (which we refer to as *objects*) over the

scalar attributes of LENGTH (in meters) and PRICE (in US Dollars).

For each object and scale, DoQ provides an empirical distribution over possible values (e.g. Figure 2) that we map into the 12 aforementioned buckets and treat it as "ground truth". We note

that DoQ itself is derived heuristically from web

text and itself contains noise; however, we use it as a starting point to evaluate the performance of

different models. Moreover, we validate our findings with transfer experiments shown in Section 6,

using DoQ to train a probe that is evaluated on the ground-truth data of Forbes and Choi (2017) and

Jianno Ni (2019).

To explore the role of context in scalar probing,

We also trained specialized probing models on a subset of DoQ data in narrow domains: MASS of

HIGHLIGHT [ann_163]: the full data distribution is needed.

PARA [para_290]: Comparing results by encoders, we see that the full data distribution is needed.

COMMENT [ann_164]: Word2Vec performs significantly worse than the

COMMENT [ann_190]: Which does perform significantly worse than the

COMMENT [ann_199]: contextual, indicating that contextual information

correct. PARA [para_333]: during pre-training improves t

PARA [para_334]: scales

HIGHLIGHT [ann_166]: Despite being weaker than BERT on down-

PARA [para_326]: stream NLP tasks, ELMo does be

COMMENT [ann_197]: Despite being weaker than BERT on down-

COMMENT [ann_198]: stream NLP tasks, ELMo does better on scalar

COMMENT [ann_199]: probing, consistent with it being better at numer-

correct. PARA [para_339]: tokenization.

PARA [para_340]: NumBERT does consistently bett...

PARA [para_416]:

HIGHLIGHT [ann_165]: and

PARA [para_342]: benchmarks such as Q/A and NLI, where Num

and Accuracy. This is in contrast to other standard

benchmarks such as Q/A and NLI, where Num

HIGHLIGHT [ann_172]: key takeaways is that the numerical representation

HIGHLIGHT [ann_173]: has a similar to the numerical representation

has an impact on scale prediction (see Figure 2

HIGHLIGHT [ann_175]: sensitive to the choice of evaluation metric. A

HIGHLIGHT [ann_192]: discussed in Section 5, we believe EMD to be the most

PARA [para_349]: robust metric a priori, but th

PARA [para_350]: the need to still examine the

robust metric a priori, but this finding highlights

the need to still examine the full range of metrics.

COMMENT [ann_200]: This is again

HIGHLIGHT [ann_178]: our analysis on Animal Masses (Table 1) show that

COMMENT [ann_201]: intuitive

can significantly improve scalar prediction, under-

scoring the importance of context. For example,

PARA [para_356]: while crane in general can c

PARA [para_357]: a piece of construction equipm

PARA [para_358]: is relevant in the animal dom

PARA [para_359]: a simpler distribution of mass

PARA [para_360]: Note that, despite signi cant

PARA [para_361]: raw numbers for each scale (ma

PARA [para_362]: relative behavior of encoders

PARA [para_363]: broadly applicable

PARA [para_364]: Transfer experiments On the F&C...

Transfer experiments On the F&C relative com-

parison task (Table 2), rgr+NumBERT performed

best, approaching the performance of using DoQ

HIGHLIGHT [ann_184]: for this task (Yang et al. 2018)

for this task (Yang et al., 2018). Scalar probes

trained with mcc perform poorly, possibly because

HIGHLIGHT [ann_185]: useful for the 8-class com

PARA [para_369]: the Amazon price dataset (Tabl

PARA [para_370]: distribution prediction task

the Amazon price dataset (Table 3) which is a full

distribution prediction task, mcc+NumBERT did

best on all three metrics. NumBERT was the bes

PARA [para_371]: for tasks. NumBERT was the bes

configurations of metric/objective, suggesting that

manipulating numeric representations can signifi-

PARA [para_374]: dev test

PARA [para_375]: mcc rgr mcc rgr

dev test

	mcc	rgr	mcc	rgr
word2vec	.40	.73	.38	.74
ELMo	.47	.71	.47	.72
BERT	.48	.71	.49	.71
NumBERT	.54	.77	.54	.76
DoQ [Elazar et. al. 2019]	-	.86	-	.87
Yang et. al. '18	-	.86	-	.87

PARA [para_379]: Table 2: Accuracy on VerbPhysi...

Table 2: Accuracy on VerbPhysics (Forbes and Choi, 2017).

PARA [para_380]: Accuracy MSE EMD

	Accuracy	MSE	EMD
Aggregate	.04	.04	.02
word2vec	.09	.23	.07
BERT	.14	.25	.07
NumBERT	.18	.27	.05

PARA [para_381]: Aggregate results on consumer price data (Jianmo Ni, 2019).

Table 3: Results on consumer price data (Jianmo Ni, 2019).

HIGHLIGHT [ann_176]: cantly improve performance on scalar prediction.

cantly improve performance on scalar prediction.

PARA [para_354]: Conclusion

PARA [para_386]: From our novel scalar probing ...

PARA [para_387]: From our novel scalar probing experiments, we

find there is a significant amount of scale informa-

tion in object embeddings, but still a sizable gap

to overcome before LMs achieve this prerequisite

of CSR. We conclude that although we observe

some non-trivial signal to extract scale information

from language embeddings, the weak signals sug-

gest these models are far from satisfying common

sense scale understanding.

PARA [para_395]: Our analysis points to improve...

Our analysis points to improvements in model-

ing context and numeracy as directions in which

progress can be made, mediated by the transfer

of scale information from numbers to nouns. The

NumBERT intervention has a measurable impact

on scalar probing results, and transfer experiments

suggest that it is an improvement. For future work

we would like to extend our models to predict

scales for sentences bearing relevant context about

scalar magnitudes, e.g. "I saw a baby elephant".

PARA [para_409]: Acknowledgments

PARA [para_406]: We want to thank Daniel Spokoy...

PARA [para_407]: We want to thank Daniel Spokoy for the idea of

using scientific notation for numbers and Jeremiah

Liu for helpful discussions on statistical distance

measures.

References

- L Baringhaus and N Henze. 2017. Cramér–von mises distance: probabilistic interpretation, confidence intervals, and neighbourhood-of-model validation. *Journal of Nonparametric Statistics*, 29(2):167–188.
- Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. 2020. Piqa: Reasoning about physical commonsense in natural language. In *Thirty-Fourth AAAI Conference on Artificial Intelligence*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Yanai Elazar, Abhijit Mahabal, Deepak Ramachandran, Tania Bedrax-Weiss, and Dan Roth. 2019. How large are lions? inducing distributions over quantitative attributes. In *Association for Computational Linguistics (ACL)*.
- Allyson Ettinger, Ahmed Elgohary, and Philip Resnik. 2016. Probing for semantic evidence of composition by means of simple classification tasks. In *Proceedings of the 1st Workshop on Evaluating Vector-Space Representations for NLP*, pages 134–139.
- Maxwell Forbes and Yejin Choi. 2017. [Verb physics: Relative physical knowledge of actions and objects](#). In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 266–276, Vancouver, Canada. Association for Computational Linguistics.
- Mor Geva, Ankit Gupta, and Jonathan Berant. 2020. Injecting numerical reasoning skills into language models. *arXiv preprint arXiv:2004.04487*.
- John Hewitt and Percy Liang. 2019. Designing and interpreting probes with control tasks. *arXiv preprint arXiv:1909.03368*.
- Julian McAuley Jianmo Ni, Jiacheng Li. 2019. Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In *Empirical Methods in Natural Language Processing (EMNLP)*.
- Bill Yuchen Lin, Seyeon Lee, Rahul Khanna, and Xiang Ren. 2020. Birds have four legs?! numersense: Probing numerical commonsense knowledge of pre-trained language models. *arXiv preprint arXiv:2005.00683*.
- Hong Liu, Mingsheng Long, Jianmin Wang, and Michael I. Jordan. 2019. Transferable adversarial training: A general approach to adapting deep classifiers. In *Proceedings of the 36th International Conference on Machine Learning*.
- Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. 2013. Distributed representations of words and phrases and their compositional-ity. In *Advances in neural information processing systems*, pages 3111–3119.
- Matthew E Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. 2018. Deep contextualized word representations. *arXiv preprint arXiv:1802.05365*.
- Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell. 2020. Information-theoretic probing for linguistic structure. *arXiv preprint arXiv:2004.03061*.
- Yossi Rubner, Carlo Tomasi, and Leonidas J Guibas. 1998. A metric for distributions with applications to image databases. In *Sixth International Conference on Computer Vision (IEEE Cat. No. 98CH36271)*, pages 59–66. IEEE.
- Xing Shi, Inkit Padhi, and Kevin Knight. 2016. Does string-based neural mt learn source syntax? In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1526–1534.
- Aaron Van Oord, Nal Kalchbrenner, and Koray Kavukcuoglu. 2016. Pixel recurrent neural networks. In *International Conference on Machine Learning*, pages 1747–1756.
- Eric Wallace, Yizhong Wang, Sujian Li, Sameer Singh, and Matt Gardner. 2019. [Do NLP models know numbers? probing numeracy in embeddings](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 5306–5314, Hong Kong, China. Association for Computational Linguistics.
- Yiben Yang, Larry Birnbaum, Ji-Ping Wang, and Doug Downey. 2018. [Extracting commonsense properties from embeddings with limited human guidance](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 644–649, Melbourne, Australia. Association for Computational Linguistics.
- Ben Zhou, Qiang Ning, Daniel Khashabi, and Dan Roth. 2020. Temporal common sense acquisition with minimal supervision. In *Association for Computational Linguistics*.

A Model Hyperparameters

Here we provide the model hyperparameters we use for reproducibility.

A.1 Probing Layer of the Scalar Probing Model

For the regression model, we use a ridge regression with regularization strength of 1. For the multi-class classification model, we use a linear classifier with a softmax activation function and regularization strength of 0.01.

Task	Metric	BERT Base	NumBERT
CoLA	Dev Acc	.745	.742
MNLI	Dev Matched Acc	.791	.789
	Dev Mismatched Acc	.795	.798
MRPC	Dev Acc	.816	.802
Squad v1	F1	.799	.789
Squad v2	Best F1	.669	.673
STS-B	Dev Pearson's r	.866	.871

Table 4: NumBERT vs BERT-base on a suite of standard NLP benchmarks.

For experiments on the narrow domains with smaller datasets, we first use PCA to reduce embeddings down to 150 dimensions before training the probing model.

A.2 NumBERT

NumBERT is pretrained on the Wikipedia and Books corpora used by the original BERT paper (Devlin et al., 2018). The BERT configuration is the same as BERT-Base (L=12, H=768, A=12, Total Parameters=110M). The language model masking is applied after WordPiece tokenization with a uniform masking rate of 15%. Maximum sequence length (number of tokens) is 128. We train with batch size of 64 sequences for 1,000,000 steps, which is approximately 40 epochs over the 3.3 billion word corpus. All the other hyperparameters and implementation details (optimizer, warm-up steps, etc.) are the same as the original BERT implementation. Table 4 shows a comparison of NumBERT vs a re-implementation of BERT-base with identical settings as above, on a suite of standard NLP benchmarks, and we conclude that the two models reach similar performance on these tasks.

B Data Statistics

Table 5 shows the statistics of 3 datasets/resources we use in this paper. For DoQ, we take the original resource and get each subset by filtering using the corresponding dimensions and/or object types (e.g. all objects, animals, product categories, etc.). Also, only objects with more than 100 values collected in the resource are used. For F&C Cleaned dataset, we use the data and the train/dev/test splits from (Elazar et al., 2019).

Dataset	Subset	#Data Samples
DoQ	all masses	76,424
	all prices	212,277
	all lengths	244,517
	animal masses	519
	product category prices	1,789
Product Price	-	1,888
F&C Cleaned	train	172
	dev	1,267
	test	1,522

Table 5: Statistics of Datasets/Resources used in our paper

C Complete Experimental Results

We model the distributions of those scalar attributes as categorical distributions over 12 categories. We first take the base-10 logarithm of all the values and then round them to the nearest integer (between -2 and 9 for all scales). We treat each integer as a bucket and use the normalized counts in each bucket as the true distribution for that scalar attribute of the object.

To explore the effect of ambiguity, we divide all the data in DoQ for each scale into 2 sets, **Unimodal** where the distribution has one well-defined peak and **Multimodal**, where multiple peaks are present. The number of peaks were identified by a simple hill-climbing algorithm.

As words often have more than one meaning in different contexts or even modifiers, their corresponding distribution from DoQ should reflect the different senses if they appeared enough in the data. When the objects are different enough (e.g. an ice-cream have mainly one meaning and its size doesn't vary much, as opposed to a truck which can be a toy truck, which is very small, or an actual vehicle, which is very big), they may have different modalities. In order to better understand our results, we wish to separate between objects of different modalities to objects with a single modality.

In order to estimate a multi-modal function, we take the bucketed DoQ distribution and smooth it into a probability density function. Then, by finding local maxima over the fitted density function, we estimate a distribution to be multi-modal if we find more than one maximum, otherwise we determine it to be a single-modal distribution.

The complete experiment results including the multimodal experiments are in Table 6.

		Accuracy			MSE			EMD			
		All	Multi	Uni.	All	Multi	Uni.	All	Multi	Uni.	
		EQUATION [ann_eqn_203]: Aggregate: .240 .250 .230									
Lengths	mcc	Aggregate	.240	.250	.230	.027	.028	.025	.077	.078	.075
		word2vec	.300	.310	.280	.026	.022	.031	.079	.074	.087
		ELMo	.430	.420	.440	.019	.020	.017	.055	.056	.053
		BERT	.420	.410	.420	.020	.021	.018	.056	.058	.054
		NumBERT	.400	.400	.410	.021	.022	.019	.052	.053	.049
	rgr	Aggregate	.240	.250	.230	.027	.028	.025	.077	.078	.075
		word2vec	.120	.120	.130	.099	.100	.097	.072	.070	.074
		ELMo	.230	.230	.240	.084	.085	.082	.072	.070	.074
		BERT	.240	.230	.240	.084	.085	.081	.072	.070	.074
		NumBERT	.220	.210	.220	.086	.088	.084	.072	.070	.074
Masses	mcc	Aggregate	.150	.150	.150	.026	.027	.024	.076	.077	.074
		word2vec	.260	.260	.260	.025	.026	.023	.082	.083	.080
		ELMo	.360	.360	.360	.021	.021	.019	.061	.062	.059
		BERT	.330	.330	.330	.021	.022	.019	.062	.063	.060
		NumBERT	.320	.320	.330	.021	.022	.019	.057	.058	.055
	rgr	Aggregate	.150	.150	.150	.026	.027	.024	.076	.077	.074
		word2vec	.200	.190	.200	.088	.090	.086	.077	.076	.080
		ELMo	.210	.200	.210	.087	.088	.085	.077	.076	.080
		BERT	.220	.210	.220	.085	.086	.084	.077	.076	.080
		NumBERT	.200	.190	.200	.088	.089	.086	.077	.076	.080
Prices	mcc	Aggregate	.240	.240	.250	.019	.021	.016	.057	.060	.054
		word2vec	.260	.250	.280	.019	.014	.024	.063	.055	.072
		ELMo	.370	.360	.380	.016	.018	.013	.051	.053	.047
		BERT	.330	.320	.330	.017	.019	.014	.054	.055	.051
		NumBERT	.320	.320	.330	.017	.019	.014	.051	.053	.048
	rgr	Aggregate	.240	.240	.250	.019	.021	.016	.057	.060	.054
		word2vec	.140	.130	.150	.090	.093	.085	.087	.084	.092
		ELMo	.210	.210	.220	.081	.083	.078	.087	.084	.092
		BERT	.190	.190	.190	.083	.085	.081	.087	.084	.092
		NumBERT	.170	.180	.170	.085	.087	.083	.087	.084	.092
Animals Masses	mcc	Aggregate	.300	.280	.330	.022	.021	.024	.059	.055	.064
		word2vec	.330	.320	.350	.021	.020	.023	.069	.066	.075
		ELMo	.430	.440	.420	.016	.015	.019	.057	.056	.059
		BERT	.410	.390	.450	.017	.016	.019	.058	.057	.060
		NumBERT	.420	.430	.410	.018	.016	.020	.053	.052	.055
	rgr	Aggregate	.300	.280	.330	.022	.021	.024	.059	.055	.064
		word2vec	.350	.350	.360	.069	.069	.069	.077	.081	.070
		ELMo	.280	.250	.330	.079	.080	.077	.077	.081	.070
		BERT	.260	.260	.240	.079	.076	.085	.077	.081	.070
		NumBERT	.230	.230	.240	.083	.081	.086	.077	.081	.070
Household Product Prices	mcc	Aggregate	.470	-	-	.010	-	-	.046	-	-
		word2vec	.510	.490	.540	.008	.008	.008	.041	.041	.041
		ELMo	.540	.520	.570	.007	.007	.007	.038	.038	.039
		BERT	.570	.560	.580	.007	.007	.007	.038	.038	.039
		NumBERT	.530	.530	.570	.007	.007	.007	.038	.038	.039
	rgr	Aggregate	.470	-	-	.010	-	-	.046	-	-
		word2vec	.450	.430	.480	.056	.058	.055	.092	.094	.090
		ELMo	.420	.400	.460	.058	.059	.057	.092	.094	.090
		BERT	.440	.420	.460	.057	.059	.055	.092	.094	.090
		NumBERT	.420	.390	.460	.060	.062	.057	.092	.094	.090

Table 6: Evaluation on all datasets.